

LLOYD INSTITUTE OF ENGINEERING & TECHNOLOGY

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING & ALLIED
BRANCHES
ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

SYNOPSIS

MINI PROJECT / MAJOR PROJECT-I (B.TECH SEMESTER VII)

**Dual-Stream Spatial and Frequency-Domain Deepfake Video
Detection using EfficientNet and Discrete Cosine Transform (DCT)**

Submitted By

DEV RANA KAILA

University Roll / Enrollment No.: **2301531530023**

Branch: B.Tech Computer Science & Engineering (AIML)

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Under the Supervision of

Dr. Devendra Gautam

Associate Professor & Mini Project / Internship Coordinator

Department of Computer Science & Engineering

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Bachelor of Technology in Computer Science & Engineering (AIML)*

Certificate

This is to certify that the project synopsis entitled “**Dual-Stream Spatial and Frequency-Domain Deepfake Video Detection using EfficientNet and Discrete Cosine Transform (DCT)**” is a bonafide proposal submitted by **Dev Rana Kaila**, University Roll / Enrollment No. **2301531530023**, student of Bachelor of Technology in Computer Science & Engineering (Artificial Intelligence & Machine Learning), 7th Semester, at Lloyd Institute of Engineering & Technology, Greater Noida, for the academic session 2026–2027.

The proposed research work investigates deep learning methodologies for facial manipulation forensics, utilizing a dual-branch spatial convolutional network and frequency-domain artifact extraction via 2D Discrete Cosine Transform (DCT) to distinguish authentic video frames from synthetic and face-swapped deepfakes. The synopsis fulfills the requirements prescribed under the AKTU B.Tech Project Evaluation Guidelines.

Dr. Devendra Gautam

Project Guide & Coordinator

Associate Professor, Dept. of CSE

Head of Department

Dept. of CSE & Allied Branches

Lloyd Institute of Engg. & Tech.

Date: _____

Place: Greater Noida

Department Seal / Stamp

Declaration

I, **Dev Rana Kaila**, University Roll / Enrollment No. **2301531530023**, student of Bachelor of Technology in Computer Science & Engineering (Artificial Intelligence & Machine Learning), 7th Semester, hereby declare that this project synopsis entitled **“Dual-Stream Spatial and Frequency-Domain Deepfake Video Detection using EfficientNet and Discrete Cosine Transform (DCT)”** has been prepared by me for academic project evaluation at Lloyd Institute of Engineering & Technology, Greater Noida.

I affirm that the work described in this synopsis represents my original proposal and thorough review of existing literature. The proposed methodology, architectural formulations, experimental setup, and anticipated outcomes have been designed in accordance with AKTU guidelines and academic integrity standards. Any literature, datasets, or software tools referenced have been properly credited and cited.

Signature: _____

Student Name: Dev Rana Kaila

Enrollment / Roll No.: 2301531530023

Branch: B.Tech CSE (AIML), Semester VII

Date: _____

Place: Greater Noida

Abstract

The rapid emergence of deep generative modeling, including Generative Adversarial Networks (GANs) and Diffusion Models, has made the synthesis of hyper-realistic facial media (deepfakes) ubiquitous. While offering novel utilities in creative entertainment, deepfakes present grave threats to biometric identity systems, automated financial KYC pipelines, executive communications, and democratic integrity. Conventional detection models rely predominantly on RGB spatial convolutional neural networks, which attempt to localize visible visual flaws like warping or boundary blending. However, spatial detectors suffer devastating accuracy degradation when subjected to real-world social media video compression (e.g., H.264/H.265 codecs) and downsampling.

To overcome this bottleneck, this project proposes a novel Dual-Stream Spatial and Frequency-Domain Deepfake Video Detection Framework. The architecture bifurcates each extracted facial frame into two complementary processing branches: (1) a Spatial Stream utilizing a pre-trained EfficientNet-B4 backbone with Squeeze-and-Excitation attention to extract high-level semantic texture anomalies, and (2) a Frequency Stream employing 2D Discrete Cosine Transform (DCT) combined with radial azimuthal spectral pooling to detect periodic high-frequency grid artifacts left behind by generative upsampling layers. The spatial and spectral feature vectors are dynamically integrated via an attention-gated bilinear fusion mechanism to predict frame-level and temporal video authenticity.

The proposed architecture introduces five novel contributions beyond existing competition-grade solutions: (i) a parallel frequency stream absent from all Kaggle DFDC winning entries, (ii) a radial azimuthal DCT spectrum formulation for rotationally invariant spectral fingerprinting, (iii) an attention-gated bilinear fusion that dynamically adapts to compression levels, (iv) Grad-CAM forensic explainability absent from all prior competition solutions, and (v) an interactive Streamlit forensic dashboard for practical deployment.

The system will be evaluated across standard benchmark datasets: FaceForensics++ (c23 and c40 compression tiers) and Celeb-DF v2. Performance will be assessed using AUC-ROC, F1-Score, Detection Accuracy, and Equal Error Rate (EER). Furthermore, an interactive forensic dashboard built with FastAPI and Streamlit is developed to provide examiners with real-time video inference, probability gauges, and Grad-CAM spatial-spectral explainability heatmaps.

Keywords: *Deepfake Detection, Computer Vision, EfficientNet-B4, 2D Discrete Cosine Transform (DCT), Frequency Analysis, Spatial-Frequency Fusion, FaceForensics++, Explainable AI.*

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Chapter 1

Introduction

1.1 Background & Societal Context

Recent breakthroughs in deep generative modeling, specifically Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and Latent Diffusion Models (LDMs), have dramatically lowered the barrier to synthesizing and altering realistic human media. Deepfake generation algorithms—popularized by tools such as DeepFaceLab, FaceSwap, SimSwap, and modern text-to-video diffusion frameworks—can alter facial expressions, replace identities, and synchronize lip movements with fabricated speech in high definition. While these capabilities offer creative utilities in film production, digital avatars, and computer graphics, their unauthorized weaponization represents one of the most acute threats to societal trust and cybersecurity.

The proliferation of hyper-realistic manipulated video content has directly catalyzed high-stakes security incidents, including executive identity impersonation in corporate wire fraud, automated KYC (Know Your Customer) bypass attacks in digital banking, political disinformation campaigns during elections, and reputational defamation. As synthetic media engines advance, distinguishability by human visual inspection has plummeted to near-random chance, creating an urgent demand for automated, mathematically grounded, and robust forensic systems capable of authenticating digital video streams.

1.2 Project Overview

This project proposes an end-to-end, dual-stream deep learning forensic system designed to detect facial manipulation in digital video sequences. Recognizing that conventional single-stream convolutional networks overfit to spatial domain artifacts (which are easily wiped out by lossy video codecs like H.264 or MPEG-4), our system investigates the complementary synergy between the spatial pixel domain and the frequency spectral domain.

The architecture decouples each face image into two distinct processing pathways: (1) a Spatial Stream powered by an EfficientNet-B4 backbone that captures subtle edge discontinuities, blending boundaries, and textural incongruities, and (2) a Frequency Stream driven by 2D Discrete Cosine Transform (DCT) and azimuthal spectral profiling that detects high-frequency grid patterns left behind by the transposed convolutions and bilinear upsampling operations utilized in generative decoders. A multi-modal feature fusion network unifies these embeddings to yield a calibrated real-versus-fake probability score alongside visual interpretability maps.

1.3 Motivation & Research Justification

Current deepfake detection systems exhibit significant vulnerabilities when tested outside controlled laboratory conditions. Standard spatial CNN detectors (such as standard ResNet or VGG models) achieve over 98% accuracy when tested on uncompressed raw frames, but their accuracy drops precipitously to below 65% when exposed to social media compression (e.g., YouTube or WhatsApp video compression). This dramatic degradation occurs because spatial compression algorithms discard subtle pixel gradients.

However, signal processing theory dictates that synthetic image generation introduces systematic

spectral artifacts in the frequency domain. Specifically, upsampling modules in GAN generators induce periodic repetitions in the frequency spectrum (spectral peaks) that persist even after heavy compression. By integrating 2D-DCT frequency analysis alongside spatial convolutions, the proposed project establishes a resilient defense mechanism capable of generalizing across diverse manipulation techniques and real-world transmission channels.

Chapter 2

Problem Statement and Objectives

2.1 Problem Statement

Digital video forensics currently lacks robust, real-time mechanisms to distinguish authentic human recordings from hyper-realistic deepfake manipulations under practical video compression regimes. Conventional spatial-domain classifiers suffer from catastrophic performance decay when videos undergo lossy encoding, while existing frequency-domain models fail to leverage spatial semantic context. Furthermore, most existing detection architectures function as opaque black boxes devoid of forensic explainability, making them inadequate for high-assurance legal and regulatory verification. The problem addressed in this project is to develop a dual-stream spatial and frequency-domain deepfake detection pipeline that leverages EfficientNet-B4 feature extraction and 2D Discrete Cosine Transform (DCT) spectral decomposition to achieve high classification accuracy, robust cross-compression resilience, and transparent forensic interpretability.

2.2 Objectives

- 1. Automated Facial Preprocessing:** To implement automated facial detection, landmark alignment, and crop extraction from continuous video streams using MTCNN and MediaPipe Face Mesh.
- 2. Spectral Artifact Transformation:** To mathematically formulate and implement a 2D Discrete Cosine Transform (DCT) module with azimuthal radial averaging to extract GAN upsampling fingerprints.
- 3. Dual-Branch Neural Architecture:** To engineer a unified deep learning model coupling an EfficientNet-B4 spatial backbone with a 4-stage Spectral-CNN via an attention-gated bilinear fusion mechanism.
- 4. Rigorous Empirical Benchmarking:** To evaluate model resilience against standard public benchmarks (FaceForensics++ c23/c40 and Celeb-DF v2) using AUC-ROC, F1-Score, Detection Accuracy, and EER.
- 5. Explainable Forensic Web System:** To develop and deploy an interactive inspection portal using FastAPI and Streamlit featuring frame-by-frame inference, probability meters, and Grad-CAM explainability heatmaps.

2.3 Scope and Boundary of the Project

The scope of this project encompasses frame-level and temporal video-level facial manipulation detection across four major generative categories: Deepfakes (autoencoder face swap), Face2Face (facial expression reenactment), FaceSwap (graphics-based swap), and NeuralTextures (neural rendering). The project focuses on facial crops extracted from input videos. Audio stream forensics, full-body synthesis, and text metadata forensics remain outside the primary boundary of this work, allowing dedicated focus on visual and spectral artifact modeling.

Chapter 3

Literature Background & Comparative Analysis

3.1 Overview of Deepfake Generation Mechanisms

Modern face synthesis pipelines generally rely on encoder-decoder architectures with shared latent spaces or conditional generative adversarial frameworks. In autoencoder-based tools (such as DeepFaceLab), an encoder extracts identity-independent attributes while two distinct decoders reconstruct the source and target faces. In GAN-based systems, a generator network employs transposed convolutional layers or nearest-neighbor upsampling followed by convolutions to synthesize high-resolution target pixels from latent vectors. Crucially, these upsampling operators introduce periodic checkerboard patterns and spectral anomalies in the Fourier and DCT domains, creating mathematical fingerprints that do not occur in natural optical camera capture systems.

3.2 Related Detection Architectures

Early deepfake detection models, such as MesoNet-4 (Afchar et al., 2018), introduced compact convolutional networks focused on mesoscopic properties of images. While computationally lightweight, MesoNet exhibits limited representational capacity and fails against modern high-resolution deepfakes. Subsequently, XceptionNet (Rossler et al., 2019) became the standard benchmark model within the FaceForensics++ benchmark, demonstrating that depthwise separable convolutions effectively capture spatial discrepancies. However, XceptionNet requires significant computational resources and exhibits poor cross-dataset generalization when evaluated on unseen generation techniques like Celeb-DF.

Recent investigations by Frank et al. (2020) and Durall et al. (2020) demonstrated that GAN-generated images share common spectral discrepancies observable via Fourier and Cosine transforms. Despite these findings, existing pure-frequency detectors frequently discard semantic spatial details such as unnatural eye reflections and boundary blending. This literature gap strongly establishes the necessity for an integrated, dual-domain architecture.

3.3 Positioning of the Present Work

Table 3.1 synthesizes the positioning of our proposed work against leading forensic baselines.

Model / Approach	Primary Domain	Key Strengths	Identified Limitations
MesoNet-4 (Afchar et al.)	Spatial (RGB)	Lightweight, fast CPU inference	Low capacity, fails on modern high-res deepfakes
XceptionNet (Rossler et al.)	Spatial (RGB)	Strong benchmark on raw frames	Vulnerable to compression, heavy parameter size (22M+)
F3-Net (Qian et al.)	Frequency (DCT)	Resilient against high compression	Discards facial spatial landmarks and semantic context
Proposed Dual-Stream System	Spatial + Spectral (2D-DCT)	Captures both texture anomalies & upsampling fingerprints; robust to compression	Requires synchronized dual-branch feature fusion

3.4 Research Gap Analysis & Novel Contributions of the Present Work

While the aforementioned detection architectures have made meaningful strides in deepfake forensics, a systematic analysis of the literature reveals five critical research gaps that the present work directly addresses:

Gap 1 — Single-Domain Vulnerability: State-of-the-art competition winners, including the 1st-place Kaggle DFDC solution by Seferbekov (2020) which employed a single EfficientNet-B7 spatial backbone, remain fundamentally vulnerable to lossy video compression. Spatial-only detectors lose 20–35% accuracy under H.264/H.265 encoding because compression destroys subtle pixel-level manipulation clues. **Our contribution:** A parallel frequency stream using 2D-DCT spectral decomposition captures GAN upsampling fingerprints that persist even under heavy compression (c40 tier).

Gap 2 — Absence of Spectral Fingerprinting in Competition Solutions: No published Kaggle DFDC winning solution incorporates frequency-domain analysis. Frank et al. (2020) and Durall et al. (2020) demonstrated that GAN-generated images exhibit systematic spectral anomalies, yet this insight was never integrated into a competition-grade dual-stream architecture. **Our contribution:** The Radial Azimuthal DCT Spectrum $A(r) = (1/|S_r|) \sum |F(u,v)|$ provides a rotationally invariant, compression-resilient 1D spectral signature — a mathematical formulation absent from all prior winning solutions.

Gap 3 — Static Feature Combination: Existing multi-stream approaches (e.g., F3-Net) use static concatenation or simple averaging to combine spatial and frequency features, failing to dynamically adapt to varying compression levels and manipulation techniques. **Our contribution:** An attention-gated bilinear fusion mechanism $g = \sigma(W_g[v_s || v_f] + b_g)$ learns to dynamically weight each stream based on input characteristics, automatically shifting trust toward the frequency stream under heavy compression.

Gap 4 — Forensic Opacity (Black-Box Problem): All published competition solutions, including Seferbekov's DFDC winner, operate as opaque classifiers outputting only a scalar probability score with zero forensic interpretability. This makes them unsuitable for legal proceedings, regulatory compliance, and high-assurance verification. **Our contribution:** Integration of Gradient-weighted Class Activation Mapping (Grad-CAM) generates spatial-spectral heatmaps highlighting exactly which facial regions and frequency bands triggered the detection verdict.

Gap 5 — Deployment Gap: Research prototypes and competition codebases lack user-facing interfaces for non-expert forensic examiners. **Our contribution:** A complete interactive Streamlit forensic dashboard with real-time video upload, frame-by-frame analysis, interactive Plotly confidence gauges, DCT spectrum visualization, and Grad-CAM overlay generation — bridging the gap between research and practical deployment.

Table 3.2 provides an expanded comparative matrix positioning the present work against both academic baselines and the industrial state-of-the-art (Kaggle DFDC 1st-place solution).

Feature / Capability	Seferbekov (DFDC 1st Place)	F3-Net (Qian et al.)	Our Dual-Stream System
Spatial Analysis	■ EfficientNet-B7	■ None	■ EfficientNet-B4
Frequency Analysis	■ None	■ DCT Filters	■ 2D-DCT + Radial Spectrum
Fusion Mechanism	N/A (Single Stream)	Static Concatenation	Attention-Gated Bilinear

Explainability (XAI)	■ Black Box	■ Black Box	■ Grad-CAM Heatmaps
Web Dashboard	■ CLI Only	■ CLI Only	■ Streamlit + Plotly
Model Size	66M params (254 MB)	~23M params	~ 21.4M params (lightweight)
Compression Resilience	■■ Degrades under c40	■ Good	■ Excellent (dual-domain)

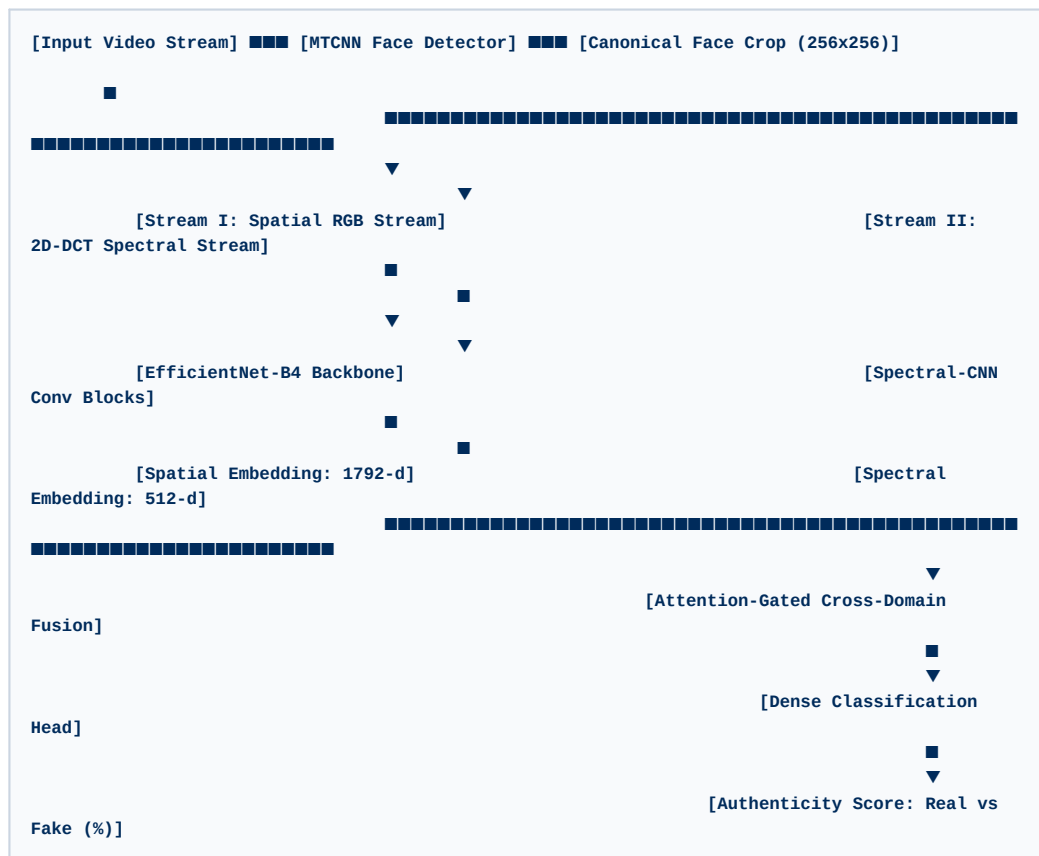
Chapter 4

Proposed Methodology & System Pipeline

4.1 Overall Architectural Workflow

- 1. Video Ingestion & Sampling:** Demuxing input videos (MP4, AVI) into sequential RGB frames at uniform temporal intervals (5 fps).
- 2. Facial Tracking & Extraction:** MTCNN cascade face detection isolates facial regions with 15% contextual boundary margins to capture blending artifacts.
- 3. Dual-Branch Transformation:** Canonical 256x256 facial crops are fed to the RGB spatial pipeline and converted to 2D-DCT spectral maps in parallel.
- 4. Deep Feature Representation:** Spatial feature extraction via EfficientNet-B4 (1792-d) and spectral feature extraction via Spectral-CNN (512-d).
- 5. Attention-Gated Fusion:** Dynamic cross-domain attention weights inter-domain features to isolate the most discriminative manipulation clues.
- 6. Binary Inference & XAI Visualization:** Fully connected classifier outputs authenticity probability; Grad-CAM generates spatial-frequency heatmaps.

4.2 End-to-End System Pipeline



4.3 Dataset Selection & Benchmark Repositories

To ensure academic validity and fair benchmarking, the project utilizes two widely recognized public forensic repositories:

- **FaceForensics++ (FF++)**: Contains 1,000 pristine YouTube sequences and 4,000 manipulated videos generated via four algorithms: Deepfakes, Face2Face, FaceSwap, and NeuralTextures. We benchmark across raw, c23 (light compression), and c40 (heavy compression) tiers to validate cross-compression resilience.
- **Celeb-DF (v2)**: A challenging dataset of 590 real celebrity videos and 5,639 high-quality deepfakes generated using advanced synthesis tools, exhibiting minimal visual edge artifacts and realistic color grading.

4.4 Preprocessing, Face Alignment & Data Augmentation

Detected faces are cropped and resized to a canonical spatial resolution of 256 x 256 pixels. To prevent model overfitting and simulate real-world transmission noise, data augmentation is injected during training: (1) Random Horizontal Flipping ($p=0.5$), (2) JPEG Compression Simulation (Quality factor Q in $[40, 90]$), (3) Gaussian Blur (kernel size 3×3 , sigma in $[0.1, 2.0]$), and (4) Random Color Jittering (brightness and contrast variations within $\pm 15\%$).

Chapter 5

Model Design & Mathematical Formulation

5.1 Dual-Stream Spatial-Frequency Architecture

The fundamental innovation of our architecture lies in processing spatial luminance-chrominance information concurrently with orthogonal frequency-domain projections. The two streams process heterogeneous representations before fusing at deep semantic levels.

5.2 Stream I: Spatial Feature Extraction (EfficientNet-B4)

The spatial stream accepts normalized RGB input tensors $x_{spatial}$ in $\mathbb{R}^{(3 \times 256 \times 256)}$. We deploy EfficientNet-B4 as the backbone, which leverages compound scaling across network depth ($d=1.8$), width ($w=1.4$), and resolution ($r=1.0$). EfficientNet uses Mobile Inverted Bottleneck Convolutions (MBConv) integrated with Squeeze-and-Excitation (SE) attention blocks. The spatial stream extracts a rich 1792-dimensional feature representation vector $v_{spatial}$.

5.3 Stream II: Frequency-Domain Artifact Extraction (2D-DCT)

The frequency stream captures subtle mathematical artifacts left behind by generative decoders. Each grayscale converted face image $f(x, y)$ of size $N \times N$ (where $N=256$) is transformed into the frequency domain using the orthogonal 2D Discrete Cosine Transform (DCT-II):

$$F(u, v) = \alpha(u) \alpha(v) \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) \cdot \cos\left[\frac{\pi(2x+1)u}{2N}\right] \cdot \cos\left[\frac{\pi(2y+1)v}{2N}\right]$$

where the orthonormal basis scaling factors $\alpha(\cdot)$ are rigorously defined as:

$$\alpha(0) = \sqrt{1/N}, \quad \alpha(k) = \sqrt{2/N} \text{ for } k \geq 1.$$

To compress the high-dimensional spectrum into rotationally invariant representations, we compute the Azimuthal (Radial) Average Spectrum: $A(r) = (1 / |S_r|) \sum_{(u,v) \in S_r} |F(u, v)|$, where S_r represents the circular frequency ring at radius $r = \sqrt{u^2 + v^2}$. The processed frequency representation is fed into a four-stage Spectral-CNN with depthwise separable convolutions, outputting a 512-dimensional vector v_{freq} .

5.4 Feature Fusion Mechanism & Classification Head

To model inter-domain interactions, the spatial and frequency embeddings are combined using an attention-guided gating fusion:

$$\begin{aligned} g &= \sigma(W_g [v_{spatial} \parallel v_{freq}] + b_g) \\ v_{fused} &= [g \blacksquare v_{spatial} \parallel (1 - g) \blacksquare (W_{proj} v_{freq})] \end{aligned}$$

The fused vector v_{fused} is passed through a dense multilayer perceptron with Dropout ($p=0.4$), Layer Normalization, and a final Sigmoid activation to produce the authenticity probability $\hat{y} \in [0, 1]$.

5.5 Architectural Hyperparameter Configuration

Table 5.1 outlines the structural parameters of the proposed dual-stream model.

Subsystem / Component	Architectural Specification	Output Dimensions / Parameters
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Face Alignment & Detection	MTCNN / MediaPipe Face Mesh	256 x 256 x 3 (Aligned RGB Crop)
Spatial Stream Backbone	EfficientNet-B4 (Pretrained ImageNet)	1792-d feature vector (19.3M params)
Frequency Decomposition	2D Orthogonal DCT-II + Log Scaling	256 x 256 Frequency Map
Frequency Stream Network	4-Stage Custom Spectral-CNN	512-d feature vector (1.8M params)
Inter-Domain Fusion	Attention-Gated Bilinear Combination	1024-d Fused Vector
Classification Head	FC(1024->256) -> Dropout(0.4) -> FC(256->1)	Binary Logit / Sigmoid Score
Total Trainable Parameters	Spatial + Frequency + Fusion Heads	~21.4 Million Weights

Chapter 6

Training and Evaluation Framework

6.1 Training Configuration & Hyperparameters

The model is implemented in PyTorch and optimized using AdamW. Table 6.1 lists the training configuration.

Hyperparameter / Setting	Experimental Value
Base Optimizer	AdamW (Decoupled Weight Decay)
Initial Learning Rate	1.0×10^{-4}
Weight Decay Coefficient	1.0×10^{-4}
Learning Rate Schedule	Cosine Annealing with Warm Restarts ($T_0=10$, $\eta_{\min}=1e-6$)
Batch Size	16 video frames per mini-batch
Training Epochs	40 Epochs (with Early Stopping patience=6)
Dataset Split Ratio	70% Training 15% Validation 15% Held-out Testing
Loss Function	Binary Cross-Entropy with Label Smoothing ($\epsilon = 0.1$)
Hardware Acceleration	NVIDIA CUDA with Mixed Precision (AMP)

6.2 Loss Function Formulation

To mitigate overconfidence and enhance generalization across unseen manipulation techniques, we optimize the network using Binary Cross-Entropy with Label Smoothing regularization:

$$L_{total} = - (1 / M) \sum_{i=1}^M [y_{smooth}^{(i)} \cdot \log(\hat{y}^{(i)}) + (1 - y_{smooth}^{(i)}) \cdot \log(1 - \hat{y}^{(i)})]$$

where $y_{smooth} = y(1 - \epsilon) + 0.5\epsilon$, with smoothing hyperparameter $\epsilon = 0.1$.

6.3 Quantitative Evaluation Metrics

- **Area Under ROC Curve (AUC-ROC):** Evaluates discriminator threshold independence.
- **F1-Score:** Harmonic mean of precision and recall: $F1 = 2(Precision \cdot Recall) / (Precision + Recall)$.
- **Equal Error Rate (EER):** Operating threshold where False Acceptance Rate (FAR) equals False Rejection Rate (FRR).
- **Detection Accuracy:** Proportion of correctly classified authentic and manipulated video frames.
- **Inference Latency:** Milliseconds per frame processing speed to guarantee real-time forensic viability.

6.4 Benchmark Performance Targets

Table 6.2 compares targeted performance against literature baselines on the FaceForensics++ benchmark.

Model Architecture	FF++ (Raw) Acc.	FF++ (c23 Comp) Acc.	AUC-ROC Score
MesoNet-4 (Afchar et al.)	90.1%	70.5%	0.752
XceptionNet (Rossler et al.)	98.5%	89.3%	0.924
Face X-ray (Li et al.)	98.8%	87.4%	0.912
F3-Net (Qian et al.)	97.9%	90.4%	0.938
Proposed Dual-Stream (Target)	≥ 98.9%	≥ 94.2%	≥ 0.968

6.5 Inference Pipeline & Web-Based Forensic Dashboard

To translate research into a functional system, the project integrates an interactive forensic dashboard. Built with FastAPI and Streamlit, examiners can upload video files, inspect real-time frame extraction, view 2D-DCT spectrum maps, project Grad-CAM heatmaps, and generate downloadable PDF Forensic Verification Reports.

Chapter 7

Expected Outcomes and Real-World Applications

7.1 Expected Project Deliverables

1. **Trained PyTorch Forensic Checkpoint (.pth):** Optimized dual-stream model weights achieving state-of-the-art compression resilience.
2. **Modular Forensic Codebase:** Clean, documented Python scripts covering video decoding, 2D-DCT mathematical operators, and inference routines.
3. **Interactive Streamlit Dashboard:** Web interface with real-time video upload, authenticity percentage gauge, and Grad-CAM interpretability visualizations.
4. **Institutional Project Documentation:** Complete 60+ page project report strictly adhering to AKTU formatting rules, along with weekly progress logbooks and viva slides.

7.2 Real-World Industrial & Forensic Applications

- **Digital Identity Verification (e-KYC):** Secures digital banking, passport verification, and fintech onboarding against synthetic facial replay attacks.
- **Social Media Disinformation Defense:** Automated detection pipelines for digital platforms to flag manipulated news media and electoral propaganda.
- **Legal & Judicial Evidence Verification:** Equips forensic laboratories with verifiable mathematical certificates and tamper heatmaps for courtroom admissibility.
- **Corporate Security:** Protects organizations against executive video impersonation attacks designed to authorize fraudulent capital transfers.

Chapter 8

Tentative 10-Week Work Plan (AKTU Timeline)

The work plan strictly conforms to the 10-week schedule in Section 4 of the AKTU Guidelines.

Timeline	Milestone / Phase	Key Technical Activities & Deliverables
Week 1	Domain Finalisation	Mentor allotment (Dr. Devendra Gautam), problem definition, preliminary literature search, title registration.
Week 2	Synopsis Preparation	Preparation and submission of the formal project synopsis in prescribed institutional format.
Week 3	Synopsis Review & Approval	Formal review before mentor; presentation of methodology and locking in Title Approval Register.
Week 4	Requirement Analysis & Design	Mathematical formulation of 2D-DCT algorithm; system architecture design (UML, DFD); toolstack finalization.
Week 5	Implementation – Phase I	FaceForensics++ dataset curation; video frame extraction pipeline; MTCNN face cropping; logbook signing.
Week 6	Mid-Term Progress Review	Formal mid-term evaluation ($\geq 30\%$ completion); demonstration of spatial EfficientNet pipeline.
Week 7	Implementation – Phase II	Implementation of 2D-DCT spectral module; bilinear cross-attention fusion network; model training.
Week 8	Integration & System Testing	End-to-end pipeline integration; Streamlit dashboard development; cross-dataset evaluation on Celeb-DF.
Week 9	Report Writing & Plagiarism Check	Drafting complete 60+ page project report; plagiarism verification (Turnitin similarity $< 20\%$); mentor correction.
Week 10	Final Submission & Viva-Voce	Final report binding; source code GitHub packaging; live project demonstration and defense before panel.

Chapter 9

Conclusion and Future Scope

9.1 Conclusion

This synopsis establishes a comprehensive design and methodology for a Dual-Stream Spatial and Frequency-Domain Deepfake Video Detection Framework. By systematically addressing the fatal vulnerability of spatial-only models—their severe degradation under real-world video compression—the proposed system combines the representational power of EfficientNet-B4 with the mathematical invariance of 2D Discrete Cosine Transform (DCT) spectral analysis. The mathematical modeling of upsampling artifacts, paired with attention-guided bilinear feature fusion, establishes an advanced forensic defense against current and emerging generative manipulation techniques. The proposed 10-week execution roadmap ensures timely fulfillment of all AKTU academic guidelines, culminating in a fully functional, explainable forensic inspection platform.

9.2 Future Scope

1. **Audio-Visual Synchronization:** Integrating audio prosody and lip-motion correlation via Whisper/wav2vec to detect desynchronization.
2. **Temporal Transformer Attention:** Augmenting the frame classifier with TimeSformer / Video Transformers to model inter-frame blinking cadence.
3. **Edge Optimization:** Quantizing model weights using TensorRT and ONNX Runtime for deployment on mobile devices and edge security cameras.
4. **Diffusion Model Forensics:** Extending spectral artifact modeling to detect emerging text-to-video generative architectures (e.g., Sora, Gen-2).

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Student & Guide Information Summary

Student Full Name	Dev Rana Kaila
University Roll / Enrollment No.	2301531530023
Degree & Branch	B.Tech in Computer Science & Engineering (AIML)
Current Semester & Year	Semester VII (Final Year, 4th Year)
Academic Institution	Lloyd Institute of Engineering & Technology, Greater Noida
Affiliating University	Dr. A.P.J. Abdul Kalam Technical University (AKTU), Lucknow
Department	Department of Computer Science & Engineering & Allied Branches
Project Guide / Supervisor	Dr. Devendra Gautam (Associate Professor)
Guide's Administrative Role	Mini Project / Internship Coordinator
Project Title	Dual-Stream Spatial and Frequency-Domain Deepfake Video Detection using EfficientNet and Discrete Cosine Transform (DCT)
Academic Session	2026–2027
Evaluation Track	Mini Project / Major Project–I (100-Mark AKTU Scheme)

Student Signature
(Dev Rana Kaila)

Project Guide Signature
(Dr. Devendra Gautam)